

URBADAPT-HEAT — Urban Heat Adaptation Framework

Probabilistic, city-level cost–benefit analysis of heat-mortality adaptation · Rome · Athens · Lisbon · Copenhagen — one pipeline, deployed across 40 European cities



`configs/{city}.yaml` · city-agnostic `template/`

Per-city parameters & policy — hazard track, income source, EWS mode, costs. One template driven by the CITY variable → drives all 10 notebooks identically



`cityheat/` `engine` · `scripts/`

Shared scientific code (nb04_masselot_main, dynamic_vulnerability, electricity_feedback, nb09...) + reproducible generator & Drive data manifests → 40-city reproducibility

Data Harmonisation

1 Setup & Loading

City configuration (YAML)
LCZ, GVI, UrbClim T2M (daily mean + max)
Population & demographics · Drive manifests

2 Harmonised Grids

Harmonise to 100 m grid (EPSG:3035)
FUA & administrative masks
Dynamic vulnerability layers

Climate Risk Model

3 Hazard & Exposure

CLIMADA hazard & exposure objects
Daily T2M (2020–2050)
Dynamic, time-varying exposure

→ Temperature hazard

4 Impact Functions

Masselot age-specific mortality ERFs
(log-linear tail extrapolation)
Burke sensitivity + cold-city Track-A/B

→ Impact functions

↓
Generic, age-resolved mortality risk

Adaptation Strategies — Each computes its own costs & benefits

5 AC Policy PRIVATE

AC coverage scenarios (SSP2)
Within-city income emulator
Efficacy-adjusted mortality reduction
CAPEX + O&M + electricity

→ Vulnerability ↓ reduces mortality given T

⚡ Side effect: Waste heat → Hazard ↑ (+ΔT)
Exports marginal deaths per °C for penalty calc

→ GROSS benefits + Costs + Marginal

6 Early Warning PUBLIC

Heat–health warning system
Three efficacy cases:
marginal / counterfactual / intermediate
Infrastructure + operational costs
25-year benefit interpolation

→ Vulnerability ↓ behavioural response

→ Benefits + Costs (25y PV)

7 Urban Trees PUBLIC

GVI cooling coefficients (ΔGVI→ΔLST, by LCZ)
 LST→T2M emulator bridge (142 cities)
Tree-maturity ramp · regional / equity allocation
CAPEX + O&M costs

→ Hazard ↓ ΔGVI→ΔLST→emulator→ΔT2M

🌱 Co-benefit: Greening → AC electricity ↓
Runtime-calibrated from JJA daily-max T2M

→ Benefits + Costs + Equity metrics

↓
Policy-specific costs, benefits & feedback parameters

Economics, Uncertainty & Synthesis — Integration & comparison

8 CBA & Optimisation

Integrate NB5 + NB6 + NB7

Waste-heat penalty: uses marginal deaths/°C → NET AC = GROSS – penalty
Vegetation–AC electricity co-benefit

25-year NPV (3% central discount)
Budget-constrained optimisation
Public vs private cost split

→ Net benefits, NPV & cost-effectiveness

9 Uncertainty

Monte-Carlo sampling over:
climate scenario & GCM
impact-function family & T-ref
adaptation efficacy
costs & discount rate
Propagated through the full pipeline

→ Probabilistic deaths avoided & CE

10 Framework Summary

Cross-pathway & cross-city synthesis
Comparative figures & tables
Ranking of adaptation options
by cost-effectiveness & equity
Reproducible per-city report

→ Decision-ready comparison